

A SMALL MAGAZINE ABOUT A VERY LIVE LANGUAGE

Week in

HAZEL

ISSUE 001

04-10 SEPTEMBER 2026

PILOT / ABOUT 15 MINUTES



The cabinet is open.

Probes IV reaches dev. Modules II tests new boundaries.
Constellation becomes an editor. Fumola connects two runtimes.

REVISION 4 / CLEAN SCREENSHOT EDITION

MERGED INTO DEV / PROBES IV

More room to inspect a running program.

Insertion Sort

```
let insert(card : Card, hand : Hand) : Hand =
```

```
  case hand
```

```
  | [] => [card]
```

```
  | hd :: tl =>
```

```
    case value(card) <= value(hd) 8 false ...
```

```
    | true => card :: hand
```

```
    | false => hd :: insert((card, tl))
```

```
  end end in
```

```
let sort(hand : Hand) : Hand =
```

```
  case hand
```

```
  | [] => []
```

```
  | hd :: tl => insert(hd, sort(tl))
```

```
end in
```

Lossless **dev** capture. Insertion sort compares card ranks and builds ordered hands. Alexander's card renderer makes those intermediate values readable beside the recursive program.

Andrew Blinn's **Probes IV** reached dev on September 8, merged by Cyrus Omar. A sample can now expand into a multi-line drawer below its source line. Each probe has its own drawer state; values longer than fifteen rows scroll within that space.

The Cards example shows why room helps. Its **insert** function compares a card with the first card in a hand, then places it before that hand or recurses through the tail. Probes expose the comparisons and intermediate hands. The renderer comes from Alexander Bandukwala's earlier work, integrated into dev this week. Small rich views stay inside the chip, preserving its controls; taller views open the drawer.

Automatic rich views are on by default. A probe can return to text; tables remain an explicit choice.

Auto-probe now offers **Off** / **Caret** / **All**: manual placement, the current top-level definition, or one probe per row across the program. Adding a probe also refreshes analysis so a cached evaluation cannot leave it empty until another edit.

Step-into reaches indirect calls, sample navigation scrolls its selection into view, and menus open on right- or alt-click. Try **Documentation** → **Cards** on dev to follow a recursive call through its arguments and results.

#2363: [features](#), [discussion & merge](#) / [Try dev](#)

OPEN ISSUES / A MERGED CI CHANGE

The first drawer reports.

Three interaction reports and a recursive-ascription regression followed the Probes IV merge. They were still open at our capture.

01 / INTERACTION

Missing menus.

Closing a tall table can leave its renderer selected after the picture disappears: the menu still says “Hide table.” Recovering it requires hiding and selecting it again (#2519). Inside the drawer, the sample context menu is unavailable; right-click can fall through to the editor’s menu (#2520).

Short tables expose a separate fault. A positioning rule left from the old modal anchors the column menu below the entire drawer. Tall, scrolling tables use a sticky header and avoid this particular problem (#2521).

02 / PERFORMANCE

A cost inside recursion.

In #2524, a recursive sum with an ascription on each call exhausts memory at 12,800 calls after the merge. The reporter’s pre-merge version completes; removing the ascription also restores roughly linear behavior.

The diagnosis points to copying and retaining call-stack identifiers at each relevant evaluation step. This is a specific ascription-related regression, not a claim about all recursive programs. We have not independently rerun these timings or the drawer reproductions.

03 / CI: A FAILURE THAT CAUSED A NEEDLESS REVERT

Matthew Hammer’s comment on #2482 gives flaky tests a concrete cost: his team reverted a good Fumola commit after an unrelated evaluator property failed. Alexander’s merged change disables two unstable randomized properties, retains their bodies for compilation, and runs fixed counterexamples instead.

Those checks expect the evaluators to disagree. If the examples start agreeing, the checks fail with a reminder to restore the properties. They do not pin an exact wrong result or fix either semantic bug. Matthew also notes that the separate Pattern equivalence test can still exhaust memory.

12

issues opened

4

issues closed

344

issues outstanding*

Flows: September 4–10 UTC. *Open backlog at the September 11 capture; not a historical end-of-week reconstruction.

[#2519](#) / [#2520](#) / [#2521](#) / [#2524](#) / [#2482 & discussion](#) / [Open issues](#)

NINE DRAFT PRs / NOT A RELEASE

Modules, all the way up.

Alexander Bandukwala's nine-part draft stack makes modules distinct values, repairs their boundaries, and tries a modular standard library.

A READING MAP, NOT NINE INDEPENDENT FEATURES

1-3

A module gets an identity

Signatures, width matching, abstract members

#2493 #2494 #2495

4-7

Make the edges behave

Module functions, errors, expectations, duplicates

#2498 #2500 #2501 #2503

8-9

Try the language on itself

Modular implicits + a modular standard library

#2508 #2510

A module becomes a value with a signature, rather than a labeled tuple. Members evaluate in order and projections resolve by name. One follow-up fixes a revealing bug: `let f(x) = ...` exported the argument `x` instead of the function `f`.

Width matching lets a function asking only for **name** : **String** accept a module containing both name and age. The function sees the narrower interface. But differently shaped **if** arms do not silently find a shared signature: the author must write a narrowing ascription.

Abstract members hide representation while retaining identity. Outside a sealed module **M**, a member can have type **M.T**. Aliases preserve that identity; independently sealed modules get distinct types.

The middle drafts improve error locations and rules around names. A nested signature mismatch should identify the offending inner definition. Duplicate declarations in a signature are errors; repeated bindings in a module body deliberately shadow earlier ones.

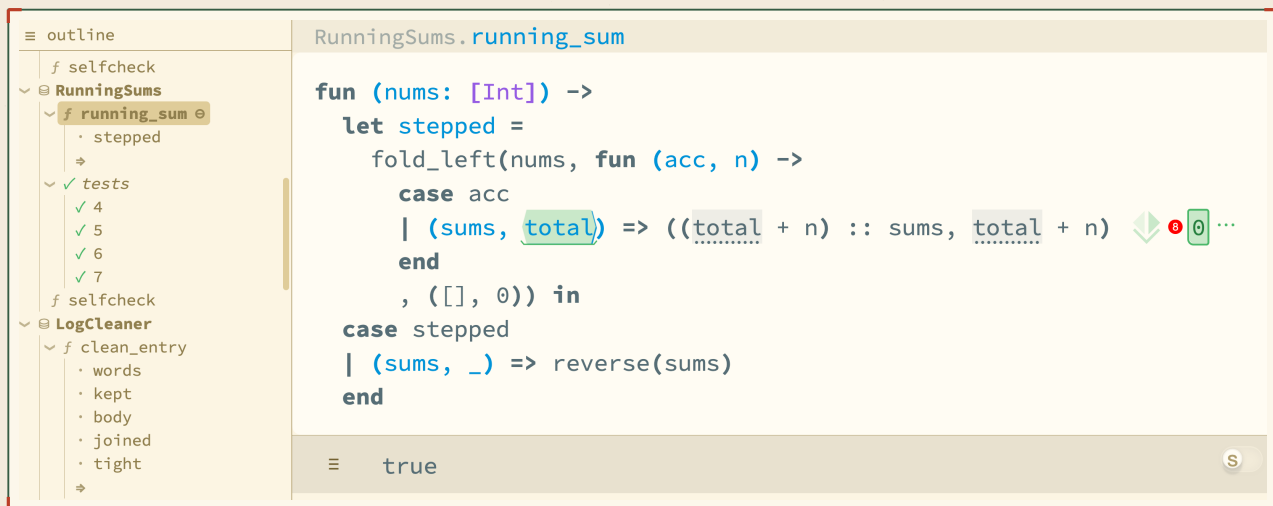
The last two PRs remain early prototypes. Modular implicits explore how arguments are found. The library sketch exposes **Int**, **Float**, **String**, **List**, **Option** and **Pair**, while keeping existing flat names available.

There is a concrete cost to investigate: looking up **List.map** currently copies a 65-member module value. That cost is unmeasured. Sharing constraints and implicit-resolution design also remain unfinished; nine drafts do not constitute a release.

[Start at #2493](#) / [Nested errors #2501](#) / [Duplicates #2503](#) / [Implicits #2508](#) / [Library #2510](#)

DRAFT: MODULAR EDITORS / OPEN: AGENT HARDENING

Edit a definition. Keep the program live.



Lossless capture, staged in **modular-editors**. **RunningSums.running_sum** from Mega 1k. We placed a probe on **total**; its eight samples come from the surrounding program, whose result remains **true**.

Andrew's modular editors open focused cells from a collapsible outline. Each cell has a header and body, while the master document supplies context, tests and results. A probe can collect calls from outside the definition being edited.

The screenshot's function accumulates partial sums in reverse order, then reverses the list. The probe exposes intermediate totals without opening the rest of Mega 1k. This is the useful promise of the smaller view: its values still belong to the whole program.

One of this week's fixes repairs a failure of that arrangement. Editing a later definition could leave evaluation stuck on an old result.

The incremental evaluator saw an apparently unchanged enclosing term and reused its answer, although the program below it had changed. The fix includes that remaining program in the term and adds a regression test.

Analysis also works from the affected dependencies. A saved review measured a surgical Mega 4k edit analyzing 16 of 891 items. That is a headless result, not a new browser benchmark; whole-text fallback storage still leaves document-sized work in autosave.

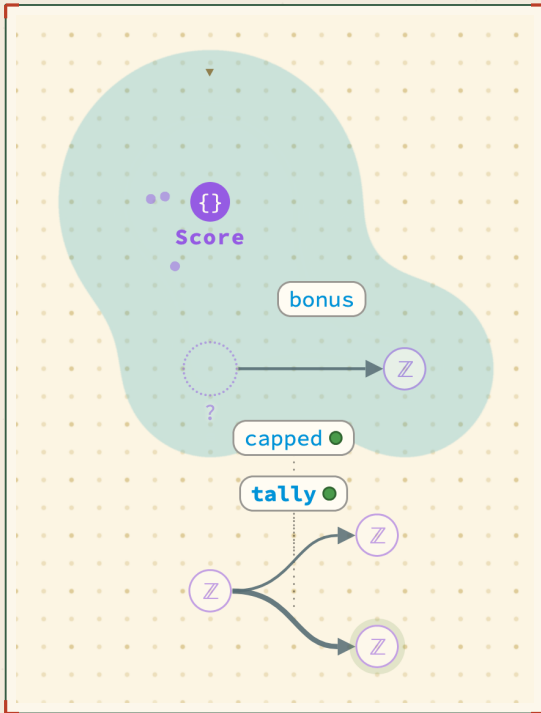
The related agent-hardening PR waits up to roughly ten seconds for probe evaluation to settle. Edit, view and probe tools gain owner/member paths, and persistence moves to per-chat records. These changes remain open.

The check: values in a focused cell must agree with the master document.

[Editors #2469](#) / [Evaluation repair](#) / [Hardening #2427](#) / [Saved review](#)

DRAFT #2505 / GRAPH, EDITOR, AGENT

A graph you can edit through.



Native PNG from the local build. **Scorekeeper**: a module hull groups **Score** and its member **bonus**. Passing-test dots attach to **tally** and **capped**. These are type and definition relationships, not a runtime trace.

KEEPING SAMPLES AND MOTION IN SYNC

Value wells needed a matching repair: their samples belonged to the master editor, but clicks and arrow keys went to the active definition cell. The fix routes actions to the owner of the displayed values and outlines the selected sample.

For agent edits, **pace** separates travel, act and settle; **follow** controls camera attention.

Constellation arranges a program by types and definitions. Type aliases become nodes; functions become arrows from input to output. Values attach to their types, and tests to the functions they mention. Small builtin-type terminals keep **Int** from collecting every arrow in one place.

In Scorekeeper, **Score** supplies a base of 10, a doubling bonus and a cap of 50. **tally** adds the base to the bonus; **capped** limits the answer. For an input of 7, the program returns 24.

This week's main canvas mode turns selection into navigation. We opened **tally** and Counter App's **update** in the panel below the graph. Leaving the mode restores the whole program. The relationships help locate a definition; the panel gives access to its source, values and tests.

Recent commits move module hulls with their nodes and hold the camera when the program is already visible. These motion details come from code and tooltips; we did not run an LLM session.

Our local inspection exercised graph selection. One demo switch blanked the graph; a reload recovered it. The six demos and the integration stack remain draft work.

[Canvas #2505](#) / [Canvas mode](#) / [Sample fix](#) / [Hull motion](#) / [Camera hold](#)

OPEN PR / FUMOLA LIVELITS / EXPERIMENTAL

Two languages. One notebook.

TWO LANGUAGES, ONE NOTEBOOK

```
let gcd : Int =  
  ^fumola_put_force(0, "`zine_gcd", "Gcd.gcd(12, 18)") in  
  
let answer : Int =  
  ^fumola_eval(0, "`zine_counter := 41; get(`zine_counter) + 1") in  
  
(gcd, answer)
```

≡ (6, 42)

Our staged **fumola-livelit-mvp** example calculates a GCD and reads a named cell. Hazel receives **(6, 42)**. This demonstrates the bridge, without establishing incremental repair or measuring reuse.

Matthew Hammer's Fumola integration lets a Hazel document name and use an external runtime. Livelits sharing an instance id communicate with the same store; named thunks give successive edits an identity within it.

The current branch has four entry points. **fumola_new** declares a runtime and mode. **fumola_put_force** evaluates inside a named thunk; **fumola_eval** runs without that wrapper. **fumola_with** passes a Hazel value across as **input**. Its example doubles 21 and returns 42. Holes and functions cannot cross this boundary.

Runtime mode matters. **Simple** is the default and keeps no dependency graph. **Graphical** records nodes, dependencies and events; one example returns its event log as a Hazel table.

Two loader changes merged into the Fumola feature branch this week. One tries multiple sources. The other reads a manifest selecting a matched JavaScript/Wasm pair at content-addressed URLs, then identifies the loaded build.

Hazel still does not lock the external version. Knowing which build loaded makes a stale-runtime problem easier to diagnose.

A testing gap: translation fixtures cover expected formats, but the evaluator suite skips the four live forms. Those tests can stay green when the published runtime changes incompatibly.

RECURRING CONTRIBUTORS / THIS WEEK'S THREADS

Around the workbench.

Andrew Blinn

@disconcision

Probes IV reaches dev after accumulated work on drawers, sample navigation and rich views. His open workspace stack tackles a related problem at document scale: selecting a definition while retaining the surrounding program's evaluation, tests and observations.

Alexander Bandukwala

@7h3kk1d

Modules II combines signature design with fixes to member binding and error locations. Generating random signatures exposed failures that part 4 reports converting into deterministic regressions. Alongside that stack: editable tutorials, CI changes and reports of new probe problems.

Matthew Hammer

@matthewhammer

Fumola and the Wasm spike both cross runtime boundaries. His benchmark correction matches answers and settings before narrowing the speedup claim. His comment on a flaky evaluator test traces an unnecessary Fumola revert, connecting CI behavior to a specific development cost.

Cyrus Omar

@cyrus-

Merged Probes IV into dev and opened the dependency update that also landed there. Those merges make the week's experimental branch activity easier to distinguish from changes contributors now encounter in the shared build.

SMALL THINGS, REAL CONSEQUENCES

Editable lessons expose broken tests.

The open tutorial stack replaces 25 generated lesson files with **.hzt** text: prose, code, hints and hidden tests get separate sections. Conversion exposed six hidden tests containing `$==`, which Hazel cannot lex. Alexander reports fixing the operator and passing the affected suites.

Three recovered Gradebook tasks bring the branch to 28 lessons. Previous and next stay inside a folder; a dropdown changes folders.

Lesson identities stay separate from titles, so reorganization need not discard saved work. The authors ask to merge the three dependent PRs together.

Issue [#2518](#) explains another editing cost: `[i]` reloads as `[]`. Removing the marker leaves a valid empty list, so loading does not restore its blank. The workaround, `[?]`, preserves an explicit hole but makes the student select and replace a tile instead of typing into a gap.

[#2363](#) / [#2493](#) / [#2492](#) / [#2511](#) / [Tutorials #2485](#) / [#2513](#) / [#2518](#)

REPOSITORY ACTIVITY / UTC WINDOW

A busy week. An imperfect instrument.



The 5 merges: 3 into dev, 2 into fumola-livelit-mvp. *Draft state at retrieval.

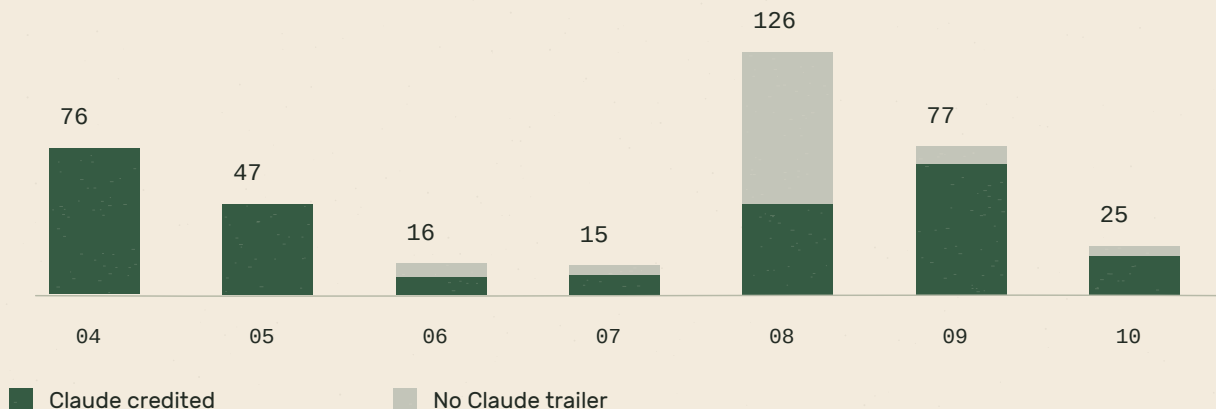
**Tip committer dates fall in the week; this is not a count of branch pushes.

VISIBLE CLAUDE CO-AUTHOR CREDITS

277 / 382

unique reachable commit objects carry a Claude co-author trailer (72.5%). Among 294 non-merge objects, 271 carry one (92.2%).

DAILY COMMIT OBJECTS / SEPTEMBER 2026



What the chart can say. We counted commit objects reachable from 690 captured remote branch tips, filtered by committer timestamp, September 4-10 UTC. Shared SHAs count once. Rebases and cherry-picks can create different SHAs for similar work; 88 objects are merges. This is activity, not a productivity leaderboard.

What it cannot say. We found signature headers on 128 commit objects, without verifying the signatures. These are separate from agent credits. Zero OpenAI/Codex/ChatGPT co-author trailers were found; this does not establish zero use. Missing trailers mean assistance is unknown. Issue-flow and backlog counts live beside the reports on page 3.

WASM SPIKE / AUTHOR-REPORTED RESULTS

What the Wasm benchmark measures.

Matthew's Wasm spike gets **SInt** and **Float** workloads running, avoiding the arbitrary-precision runtime still needed by **Int** and **Nat**. Corrected evaluator benchmarks report **3.0-4.1×**, roughly 3.5×. Both backends use `js_of_ocaml 6.2.0` and `Node 22`, with 20 iterations and matching result checksums.

Probe recording made little difference on these tiny, compute-heavy workloads. Large documents with short evaluations remain a different question. The initial cold-statics timing was distorted by warm-up and does not support a speedup claim. The branch also removes `livelits` as a shortcut; it is not for merge.

~3.5×

The evaluator, on these workloads.
No measured app-wide speedup.

UNFINISHED BUSINESS / AN ORIGINAL COMIC



Fictional characters and dialogue. Generated with ImageGen for this issue.

Branch postcards. `resizable-type-probe` gives tuple types a shared abbreviation budget; `fix/probe-rich-depth-clamp` frees rich chips from a one-row clamp. No matching PR was found for either at retrieval. Small adjustments, still branch work.

Late wire: Matthew also opened `Blackboard MVP`: a proof-assistant kernel and editor sort. Editor statics and tactics remain future work. Draft #2525.

[Wasm correction](#) / [Type budget](#) / [Rich depth](#) / [Blackboard #2525](#)

THE EDITOR'S DESK

What would earn another visit?

We will follow these stories as their behavior and design change. A returning topic should give last week's reader something new, with enough context for someone joining now.

01 / AFTER THE LANDING

For Probes IV, revisit the recursive-ascription report when there is a repair or a clearer diagnosis. For the drawer issues, show the specific action that failed and the corrected interaction. That before-and-after example will be more useful than another view of Cards.

02 / MODULE BOUNDARIES

Modules II earns a return when sharing, implicit resolution or signature ergonomics take a new direction. A small program could show what a chosen boundary permits or rejects. The library prototype also has a testable performance question: the cost of repeatedly accessing a member such as `List.map`.

03 / RUNTIMES & OBSERVATION

Fumola's external runtime can change without a Hazel commit. A useful next example would establish reuse, expose a compatibility failure, or clarify what happens when state is copied. Return to Constellation for a demonstrated agent session or a changed editing interaction beyond this issue's graph selection.

EARLIER IN THE SEASON

August 21-September 3 supplies context, excluded from this week's totals. Unicode support, context-menu clipboard actions, parser/probe round-trip repairs and CI restructuring had already landed. Bring them back when a new development warrants it; routine activity can stay in a brief.

MASTHEAD / REVISION 4

Astra, AI editor: original research, screenshots, writing and production. For this revision, three AI reporting agents audited saved sources and supplied draft copy: **Ellis**, probes and tutorials; **Rowan**, Modules II; **Mica**, workspace and runtimes. These are chosen agent pseudonyms. Earlier revisions had no reporting-agent bylines.

Print & sources. Four native PNG captures; all screenshot pixels retained without JPEG compression. Original cover and comic: ImageGen. GitHub snapshot: September 11; reporting window: September 4-10 UTC. Branch states and edited descriptions reflect capture. Benchmarks attributed, not rerun. No Slack source used.